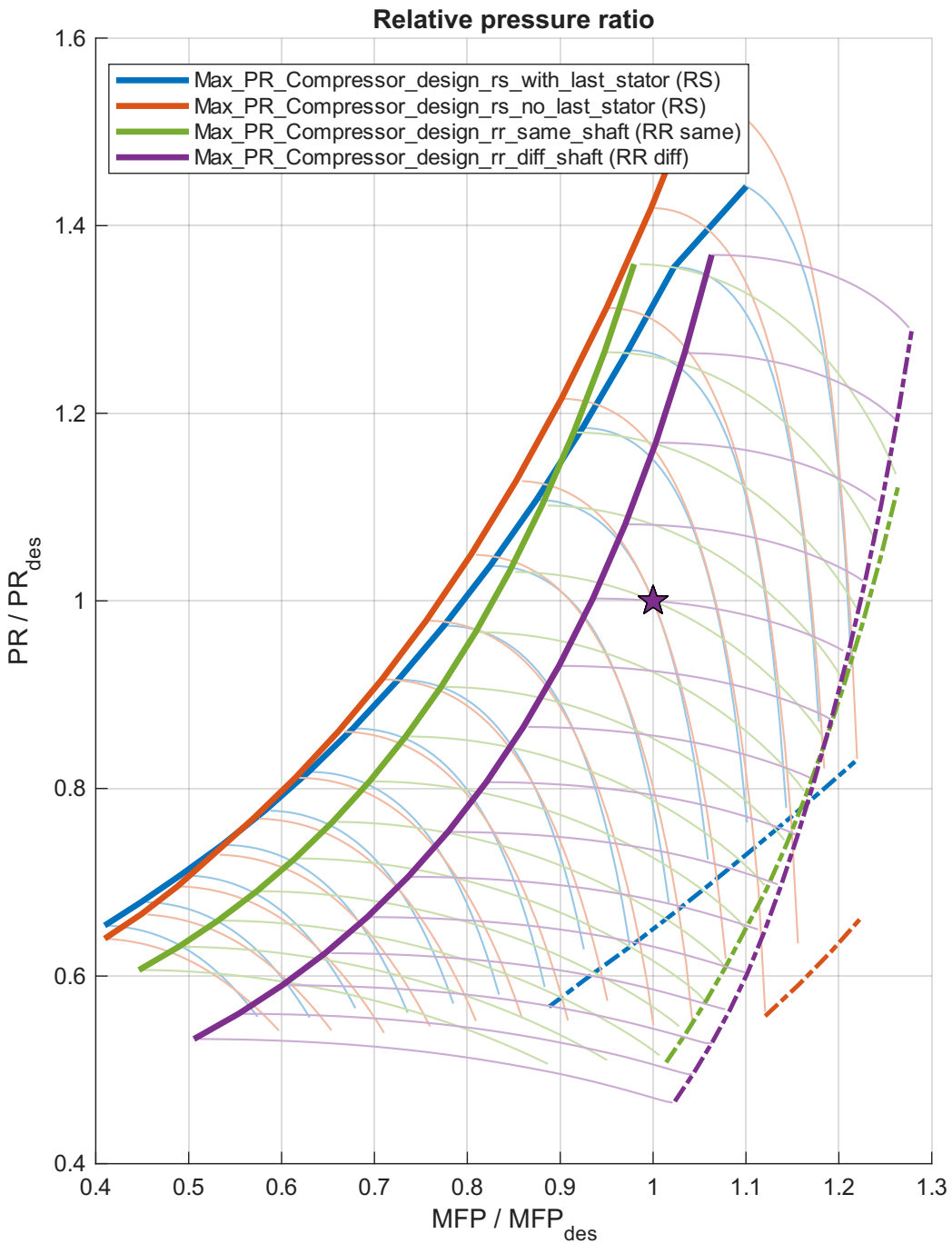
Comparison

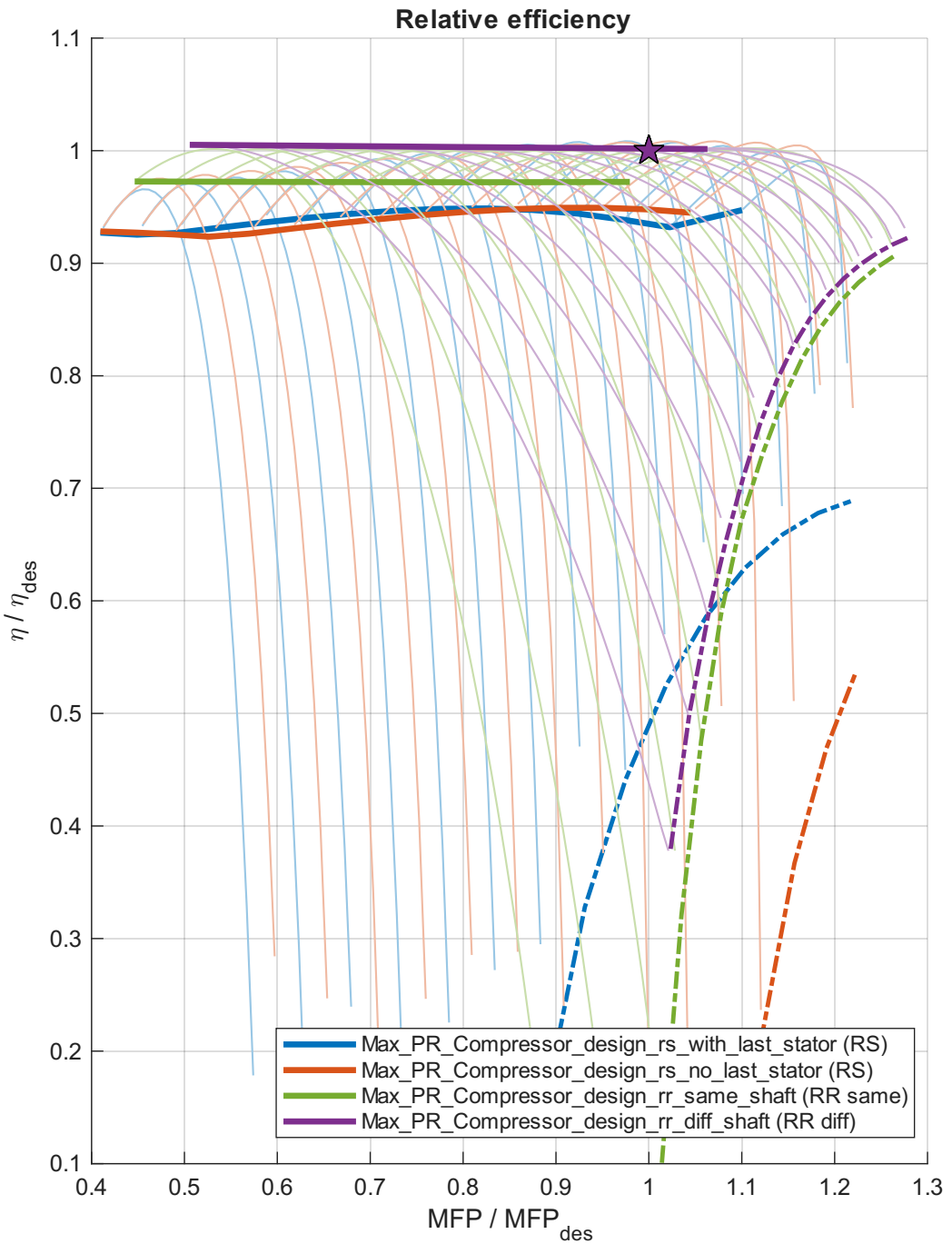
Runs:
  1. Max_PR_Compressor_design_rs_with_last_stator (RS) - Rotor-Stator - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_PR_Compressor_design_rs_with_last_stator.mat
  2. Max_PR_Compressor_design_rs_no_last_stator (RS) - Rotor-Stator - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_PR_Compressor_design_rs_no_last_stator.mat
  3. Max_PR_Compressor_design_rr_same_shaft (RR same) - Rotor-Rotor same shaft - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_PR_Compressor_design_rr_same_shaft.mat
  4. Max_PR_Compressor_design_rr_diff_shaft (RR diff) - Rotor-Rotor different shaft - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_PR_Compressor_design_rr_diff_shaft.mat

Line styles on the graphs: solid = first stability limit, dash-dot = choke line, star = design point, light lines = speed lines.

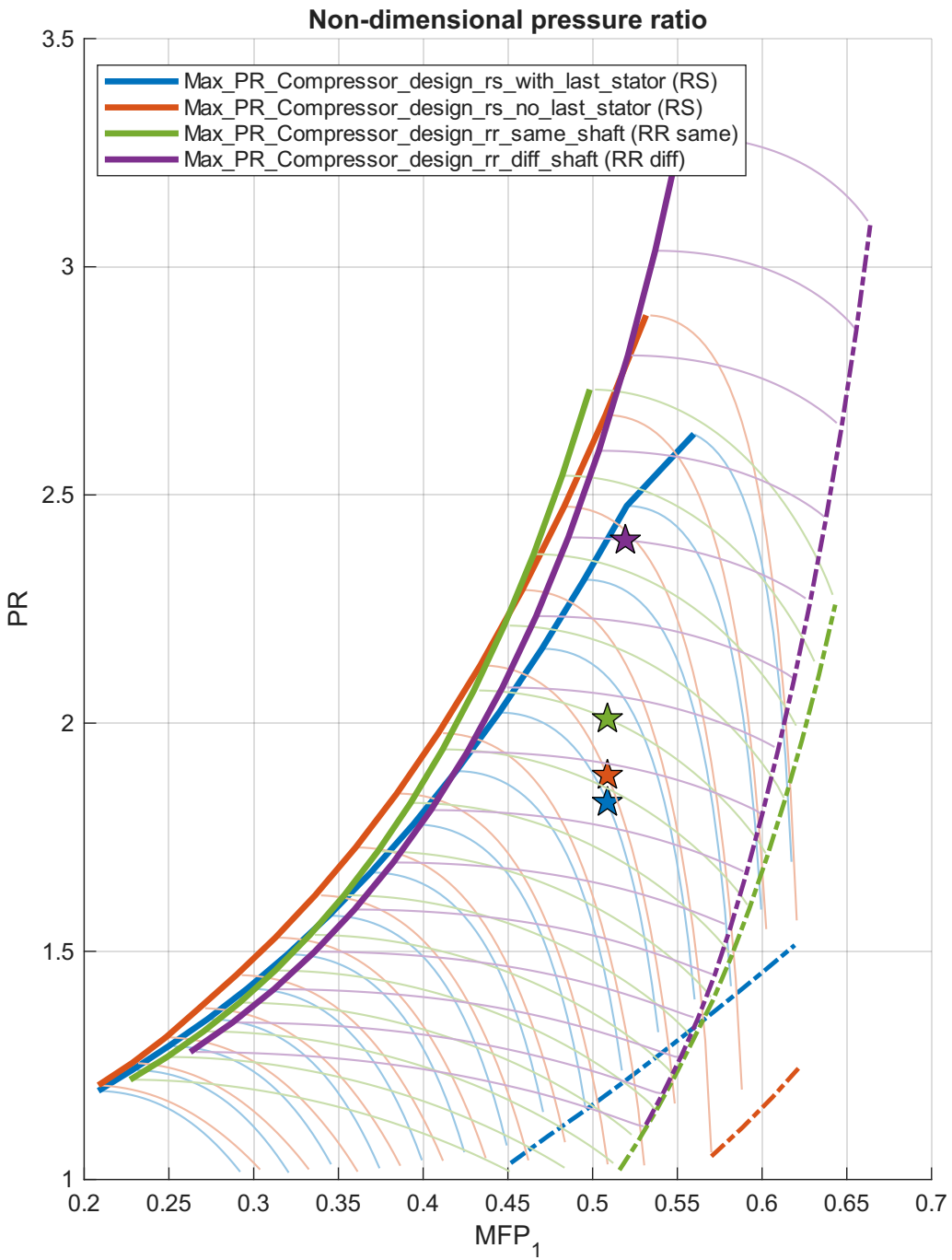
Label	Configuration	Source	PR_des	eta_des	First limit (N=1)	SM [%]	Choke margin [%]	Range [%]	Koch ratio (max)	Koch applicability	U/(2a_p)
Max_PR_Compressor_design_rs_with_la...	Rotor-Stator	Max_PR_Compressor_design_rs_with_la...	1.8264	0.7804	peak	26.7	6.2	17.6	0.731	indicative	0.406
Max_PR_Compressor_design_rs_no_last...	Rotor-Stator	Max_PR_Compressor_design_rs_no_last...	1.8855	0.8255	peak	32.3	8.3	21.3	0.731	indicative	0.405
Max_PR_Compressor_design_rr_same_sh...	Rotor-Rotor same shaft	Max_PR_Compressor_design_rr_same_sh...	2.0095	0.917	peak	22	18.5	28.7	0.891	not recommended	0.413
Max_PR_Compressor_design_rr_diff_sh...	Rotor-Rotor different shaft	Max_PR_Compressor_design_rr_diff_sh...	2.401	0.9239	peak	7.2	21	22.7	1.028	not recommended	0.331

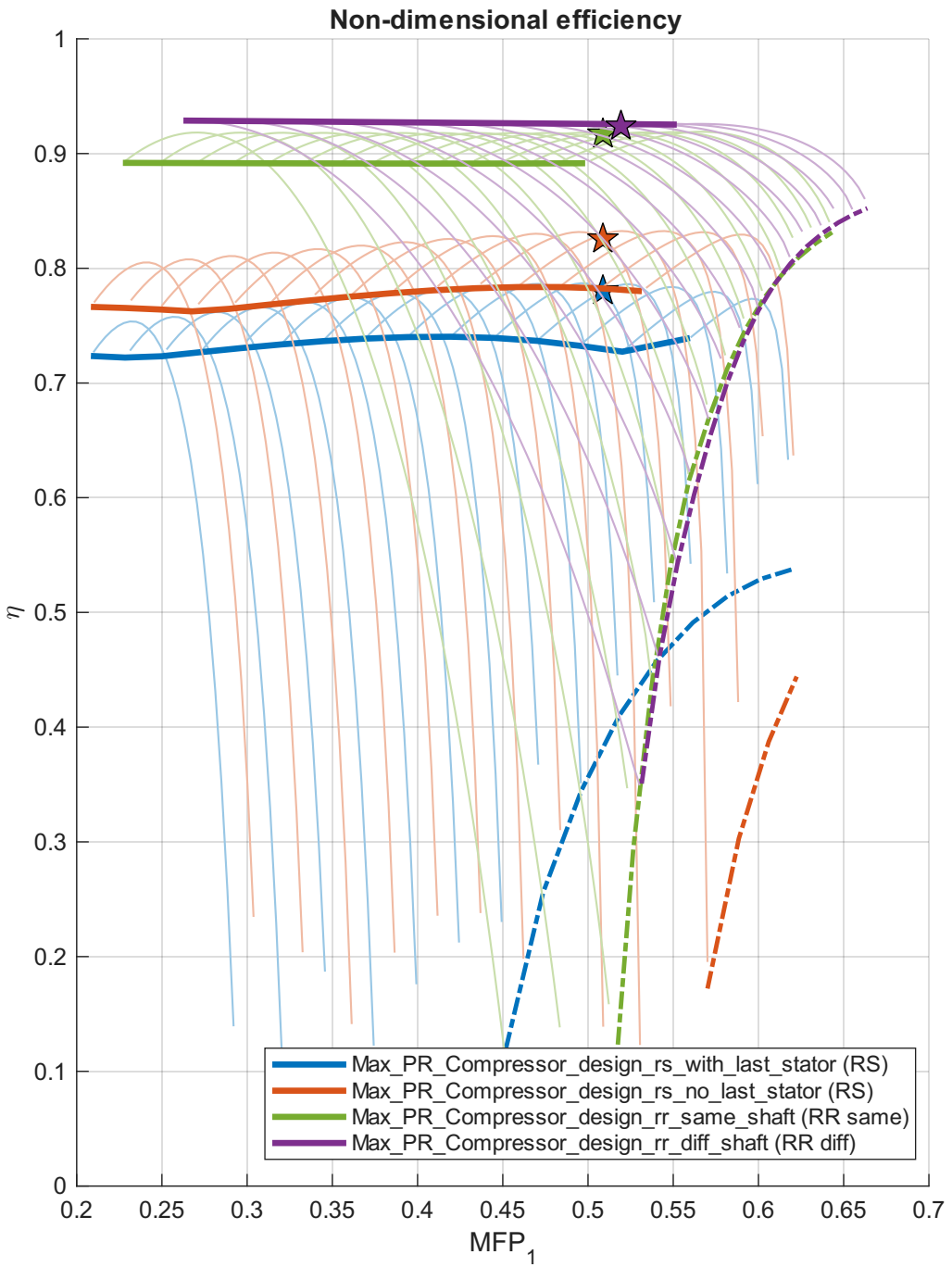
Relative comparison





NonDimensional comparison





Inputs: Max_PR_Compressor_design_rs_with_last_stator (RS)

Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_PR_Compressor_design_rs_w...		
alpha1	38.4033	deg	.mat: Max_PR_Compressor_design_rs_w...
n_stages	2	-	.mat: Max_PR_Compressor_design_rs_w...
A_vec	0.00360564 0.00323586 0.00290087 0....	m^2	.mat: Max_PR_Compressor_design_rs_w...
beta1_dash	51.8472	deg	.mat: Max_PR_Compressor_design_rs_w...
beta2_dash	34.8686	deg	.mat: Max_PR_Compressor_design_rs_w...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rs_w...
w_loss_rotor	0.140799	-	.mat: Max_PR_Compressor_design_rs_w...
deviation	3.53466	deg	.mat: Max_PR_Compressor_design_rs_w...
incidence	6.87796	deg	.mat: Max_PR_Compressor_design_rs_w...
deflection	20.3219	deg	.mat: Max_PR_Compressor_design_rs_w...
w_loss_stator	0.115656	-	.mat: Max_PR_Compressor_design_rs_w...
U	315.356	m/s	.mat: Max_PR_Compressor_design_rs_w...
sigma_rotor	1.97032	-	.mat: Max_PR_Compressor_design_rs_w...
sigma_stator	1.97032	-	.mat: Max_PR_Compressor_design_rs_w...
chord	0.01	m	.mat: Max_PR_Compressor_design_rs_w...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	135	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_PR_Compressor_design_rs_with_last_stator (RS)

```
Rotor-Stator  |  PR_des = 1.8264, eta_des = 0.7804  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = peak at m = 0.4372 kg/s (PR 2.023); stall margin SM = 26.7 %
   peak of the speed line (zero slope): m = 0.4372 kg/s | Howell stall onset: m = 0.3915 kg/s
   Koch stage limit: not reached on the computed part of the line
CHOKE (design speed): m_choke = 0.5309 kg/s (Stator 2 exit (absolute)) -> choke margin 6.2 %, operating range (m_choke - m_lim)/m_choke = 17.6 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.406 -> surge if the installation has V_p/(A_c L_c) > 3.88 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 2 (omega/omega_des = 1.57), lowest de Haller Stator 2 (0.55). Details: Tables > Blade
      rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: none to draw. No speed line reaches Ch_ef/Ch_max = 1 before the peak or Howell limit, so the stage stays below its Koch limit over the
      whole stable range.
   why: Stage 1 (R1+S1): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2+S2): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): OK (Ch_ef/Ch_max = 0.73 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.66 (stage L/g2 2.41 -> 1.60): R1 1.97 -> 1.31, S1 1.97 -> 1.31; re-run the design script
      with it (profile loss changes)
     - or RAISE the stage static-pressure rise by up to about 16% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
  Stage 2 (R2+S2): OK (Ch_ef/Ch_max = 0.73 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.66 (stage L/g2 2.41 -> 1.58): R2 1.97 -> 1.29, S2 1.97 -> 1.29; re-run the design script
      with it (profile loss changes)
     - or RAISE the stage static-pressure rise by up to about 17% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
```

Inputs: Max_PR_Compressor_design_rs_no_last_stator (RS)

Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_PR_Compressor_design_rs_n...		
alpha1	38.4033	deg	.mat: Max_PR_Compressor_design_rs_n...
n_stages	2	-	.mat: Max_PR_Compressor_design_rs_n...
A_vec	0.00360564 0.00323586 0.00290087 0....	m^2	.mat: Max_PR_Compressor_design_rs_n...
beta1_dash	51.8472	deg	.mat: Max_PR_Compressor_design_rs_n...
beta2_dash	34.8686	deg	.mat: Max_PR_Compressor_design_rs_n...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rs_n...
w_loss_rotor	0.140799	-	.mat: Max_PR_Compressor_design_rs_n...
deviation	3.53466	deg	.mat: Max_PR_Compressor_design_rs_n...
incidence	6.87796	deg	.mat: Max_PR_Compressor_design_rs_n...
deflection	20.3219	deg	.mat: Max_PR_Compressor_design_rs_n...
w_loss_stator	0.115656	-	.mat: Max_PR_Compressor_design_rs_n...
U	315.356	m/s	.mat: Max_PR_Compressor_design_rs_n...
sigma_rotor	1.97032	-	.mat: Max_PR_Compressor_design_rs_n...
sigma_stator	1.97032	-	.mat: Max_PR_Compressor_design_rs_n...
chord	0.01	m	.mat: Max_PR_Compressor_design_rs_n...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	135	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_PR_Compressor_design_rs_no_last_stator (RS)

```
Rotor-Stator  |  PR_des = 1.8855, eta_des = 0.8255  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = peak at m = 0.4263 kg/s (PR 2.127); stall margin SM = 32.3 %
   peak of the speed line (zero slope): m = 0.4263 kg/s | Howell stall onset: m = 0.3900 kg/s
   Koch stage limit: not reached on the computed part of the line
CHOKE (design speed): m_choke = 0.5416 kg/s (Rotor 2 exit (relative)) -> choke margin 8.3 %, operating range (m_choke - m_lim)/m_choke = 21.3 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.405 -> surge if the installation has V_p/(A_c L_c) > 3.89 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 1 (omega/omega_des = 1.58), lowest de Haller Rotor 1 (0.54). Details: Tables > Blade
      rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: none to draw. No speed line reaches Ch_ef/Ch_max = 1 before the peak or Howell limit, so the stage stays below its Koch limit over the
      whole stable range.
   why: Stage 1 (R1+S1): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2 only): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): OK (Ch_ef/Ch_max = 0.73 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.66 (stage L/g2 2.41 -> 1.60): R1 1.97 -> 1.31, S1 1.97 -> 1.31; re-run the design script
      with it (profile loss changes)
     - or RAISE the stage static-pressure rise by up to about 16% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
  Stage 2 (R2 only): OK (Ch_ef/Ch_max = 0.71 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.63 (stage L/g2 2.41 -> 1.51): R2 1.97 -> 1.24; re-run the design script with it (profile
      loss changes)
     - or RAISE the stage static-pressure rise by up to about 19% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
```

Inputs: Max_PR_Compressor_design_rr_same_shaft (RR same)

Variable	Value	Unit	Source
Configuration	Rotor-Rotor same shaft		
Input source	.mat: Max_PR_Compressor_design_rr_s...		
alpha1	38.4033	deg	.mat: Max_PR_Compressor_design_rr_s...
n_stages	2	-	.mat: Max_PR_Compressor_design_rr_s...
A_vec	0.00360564 0.00323629 0.00313338	m^2	.mat: Max_PR_Compressor_design_rr_s...
beta1_dash	51.8472	deg	.mat: Max_PR_Compressor_design_rr_s...
beta2_dash	34.8686	deg	.mat: Max_PR_Compressor_design_rr_s...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rr_s...
w_loss_rotor	0.141203	-	.mat: Max_PR_Compressor_design_rr_s...
beta3_dash	38.2935	deg	.mat: Max_PR_Compressor_design_rr_s...
beta4_dash	-12.5679	deg	.mat: Max_PR_Compressor_design_rr_s...
deviation_1	3.53466	deg	.mat: Max_PR_Compressor_design_rr_s...
deviation_2	9.07861	deg	.mat: Max_PR_Compressor_design_rr_s...
incidence_1	6.87796	deg	.mat: Max_PR_Compressor_design_rr_s...
incidence_2	0.109785	deg	.mat: Max_PR_Compressor_design_rr_s...
deflection_1	20.3219	deg	.mat: Max_PR_Compressor_design_rr_s...
deflection_2	41.8926	deg	.mat: Max_PR_Compressor_design_rr_s...
w_loss_rotor2	0.0800466	-	.mat: Max_PR_Compressor_design_rr_s...
U	315.356	m/s	.mat: Max_PR_Compressor_design_rr_s...
sigma_1	1.97032	-	.mat: Max_PR_Compressor_design_rr_s...
sigma_2	1.42267	-	.mat: Max_PR_Compressor_design_rr_s...
chord	0.01	m	.mat: Max_PR_Compressor_design_rr_s...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	141	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_PR_Compressor_design_rr_same_shaft (RR same)

```
Rotor-Rotor Same Shaft  |  PR_des = 2.0095, eta_des = 0.9170  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = peak at m = 0.4226 kg/s (PR 2.072); stall margin SM = 22.0 %
   peak of the speed line (zero slope): m = 0.4226 kg/s | Howell stall onset: m = 0.3899 kg/s
   Koch stage limit: not reached on the computed part of the line
CHOKE (design speed): m_choke = 0.5926 kg/s (Rotor 1 exit (relative)) -> choke margin 18.5 %, operating range (m_choke - m_lim)/m_choke = 28.7 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.413 -> surge if the installation has V_p/(A_c L_c) > 3.75 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Rotor 1 (omega/omega_des = 1.53), lowest de Haller Rotor 1 (0.53). Details: Tables > Blade
      rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 4 of 15 speed lines (N/N_des 1.05 to 1.20).
   why: R2: effective dynamic-head factor F_ef = 0.25, outside 0.39-1.11. The upstream wakes enter in the same rotating frame (same shaft), so they
      are not re-energized; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: LOW MARGIN (0.89, target 0.85)
     - RAISE the solidity x1.15 (stage L/g2 2.13 -> 2.45): R1 1.97 -> 2.26, R2 1.42 -> 1.63; re-run the design script with it (profile loss changes)
      [solidity above 2: outside the Carter/NACA-65 range of the design scripts]  [L/g2 outside Koch's data 0.6-2.2]
     - or LOWER the stage static-pressure rise by about 5% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.82
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.84
     - R2: effective dynamic-head factor F_ef = 0.25 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
      1 the ratio would be 0.70 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: F_ef = [1.08 0.25] (data 0.39-1.11) -> indicative only
```

Inputs: Max_PR_Compressor_design_rr_diff_shaft (RR diff)

Variable	Value	Unit	Source
Configuration	Rotor-Rotor different shaft		
Input source	.mat: Max_PR_Compressor_design_rr_d...		
alpha1	22.2856	deg	.mat: Max_PR_Compressor_design_rr_d...
n_stages	2	-	.mat: Max_PR_Compressor_design_rr_d...
A_vec	0.00299142 0.00268389 0.00253739	m^2	.mat: Max_PR_Compressor_design_rr_d...
beta1_dash	45.3733	deg	.mat: Max_PR_Compressor_design_rr_d...
beta2_dash	17.2902	deg	.mat: Max_PR_Compressor_design_rr_d...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rr_d...
w_loss_rotor	0.118639	-	.mat: Max_PR_Compressor_design_rr_d...
beta3_dash	38.2869	deg	.mat: Max_PR_Compressor_design_rr_d...
beta4_dash	-17.8442	deg	.mat: Max_PR_Compressor_design_rr_d...
deviation_1	4.99536	deg	.mat: Max_PR_Compressor_design_rr_d...
deviation_2	8.53425	deg	.mat: Max_PR_Compressor_design_rr_d...
incidence_1	5.50411	deg	.mat: Max_PR_Compressor_design_rr_d...
incidence_2	3.75221	deg	.mat: Max_PR_Compressor_design_rr_d...
deflection_1	28.5918	deg	.mat: Max_PR_Compressor_design_rr_d...
deflection_2	51.349	deg	.mat: Max_PR_Compressor_design_rr_d...
w_loss_rotor2	0.110832	-	.mat: Max_PR_Compressor_design_rr_d...
U1	257.281	m/s	.mat: Max_PR_Compressor_design_rr_d...
U2	334.465	m/s	.mat: Max_PR_Compressor_design_rr_d...
sigma_1	1.96031	-	.mat: Max_PR_Compressor_design_rr_d...
sigma_2	1.95306	-	.mat: Max_PR_Compressor_design_rr_d...
chord	0.01	m	.mat: Max_PR_Compressor_design_rr_d...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	143	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_PR_Compressor_design_rr_diff_shaft (RR diff)

```
Rotor-Rotor Different Shaft  |  PR_des = 2.4010, eta_des = 0.9239  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = peak at m = 0.4676 kg/s (PR 2.407); stall margin SM = 7.2 %
   peak of the speed line (zero slope): m = 0.4676 kg/s | Howell stall onset: m = 0.3665 kg/s
   Koch stage limit: m = 0.5086 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.6050 kg/s (Rotor 1 exit (relative)) -> choke margin 21.0 %, operating range (m_choke - m_lim)/m_choke = 22.7 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.331 -> surge if the installation has V_p/(A_c L_c) > 5.85 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Rotor 1 (omega/omega_des = 1.06), lowest de Haller Rotor 1 (0.63). Details: Tables > Blade
      rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 15 of 15 speed lines (N/N_des 0.50 to 1.20).
   why: R2: effective dynamic-head factor F_ef = 0.35, outside 0.39-1.11. The upstream wakes enter in a frame turning the same way at a similar
      speed; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: BEYOND THE LIMIT AT DESIGN (1.03)
     - solidity alone cannot reach the target (L/g2 > 2.8 would be needed)
     - or LOWER the stage static-pressure rise by about 17% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.94
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.96
     - R2: effective dynamic-head factor F_ef = 0.35 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
      1 the ratio would be 0.74 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: F_ef = [0.96 0.35] (data 0.39-1.11) -> indicative only
```